The Plant Doctor · Test cases

Automation vs AI activity · ESP32 (Coggy) + web app. Tick Pass or Fail and note anything odd. Run in Chrome or Edge with Bluetooth on, using the downloaded ThePlantDoctor.html (Bluetooth does not work inside the artifact preview).

Tester: _____ Table: _____ Date: _____ ESP32 sketch: PlantDoctor_Coggy.ino

A. Hardware and sensors (Arduino Serial Monitor, 115200 baud)

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-01	**Board boots and advertises**	Upload PlantDoctor_Coggy.ino. Press EN/reset. Open Serial Monitor.	"Advertising as PlantDoctor-T1" is printed. A JSON line appears every second.			
TC-02	**DHT-11 reading**	Read t and h in the JSON. Breathe on or hold the sensor for 20 s.	t and h are numbers, not null. Temperature and humidity rise. Values update about every 2 s.			
TC-03	**Soil sensor dry**	Keep the soil probe in dry soil or in air.	"sd":1. If it shows 0, adjust the module pot or flip SOIL_DRY_LEVEL.			
TC-04	**Soil sensor wet**	Dip the probe in wet soil or a cup of water.	"sd":0 within 1 to 2 s.			
TC-05	**Rain sensor**	Splash a few drops on the rain plate, then wipe it dry.	"rn":1 when wet, "rn":0 when dry. Flip RAIN_WET_LEVEL if reversed.			
TC-06	**LDR**	Cover the LDR with a hand, then uncover it under room light.	"dk":1 when covered, "dk":0 in light. Adjust the pot if it never changes.			
TC-07	**Boot with LDR covered (GPIO 12 check)**	Cover the LDR and power-cycle the board.	Board still boots. If it does not, move LDR to GPIO 34 or 35 and change LDR_PIN.			
TC-08	**Pump and LED outputs**	After connecting the web app (section B), go to AI mode and press Turn on, then Turn off, for Pump and Lights.	Pump runs and stops through the L298N (ENA jumper removed, IN2 on GND). LED on GPIO 23 lights. Serial shows CMD: PUMP:1 and so on.			

B. Bluetooth connection (Chrome or Edge, downloaded HTML file)

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-09	**Find Coggy**	Open ThePlantDoctor.html, start a session, press Connect Coggy.	The picker lists PlantDoctor-T1. After pairing the button turns green with the device name.			
TC-10	**Sensor bar only when connected**	Look under the menu before and after connecting.	Before: a "Connect Coggy" strip, no readings. After: 7 tiles with live values and a green dot.			
TC-11	**Live readings match hardware**	Change soil, rain and LDR as in section A.	Tiles flash and change within 1 to 2 s, matching the Serial Monitor.			
TC-12	**Disconnect safety**	Turn the pump on in live mode, then switch off laptop Bluetooth.	Toast "Coggy disconnected". The ESP32 stops the pump and lights and advertises again.			
TC-13	**Reconnect**	Press Connect Coggy again.	Readings resume; no page reload needed.			
TC-14	**Demo Coggy (no kit)**	Disconnect, press "No kit? Use Demo Coggy".	Demo panel opens; sensor bar shows "Demo Coggy". Buttons change soil, rain and light.			

C. Session, landing page and saved data

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-15	**Landing page**	Open the file. Scroll the page.	Logo, favicon, hero field animation, sections, footer show. Phone width has no sideways scroll.			
TC-16	**Details form validation**	Press Get started, leave School empty, press Start farming.	Error "Fill in school, grade and table name to begin."			
TC-17	**Start session**	Fill school, grade, table (team and farm optional), submit.	Dashboard opens with the guide. Table and school show in the menu chip and footer.			
TC-18	**No data lost on refresh**	Add a rule, save moments on Day 1, then refresh the page.	Same tab opens with rules, diary and progress intact. Landing shows "Session saved for ...".			
TC-19	**End session**	Press End session, then End and clear.	All data cleared; landing page shown; pump and lights sent OFF.			

D. Automation mode

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-20	**Add a ready rule**	Tap "Pump 6:00 to 6:30 AM" and "Lights on when dark".	Two rules appear numbered 1 and 2 with ON/OFF switches.			
TC-21	**Build a custom rule**	IF Temperature is above 30 °C AND Soil is dry THEN Pump ON. Press Add this rule.	Rule appears in plain words. Duplicate rules are refused.			
TC-22	**Rule preview**	With Coggy connected, change the sensors.	"What Coggy would do right now" updates and the firing rule is outlined.			
TC-23	**Stop rule beats start rule**	Add "Stop pump in rain". Make soil dry and rain sensor wet.	Preview shows Pump OFF with the stop rule as the reason.			
TC-24	**Disable / delete rule**	Switch a rule off, then delete it.	Disabled rule is faded and ignored; deleted rule is gone after refresh too.			
TC-25	**Go live needs a rule**	Delete all rules; look at Go live.	Go live is disabled with "Add at least one rule first."			

E. Live mode (Automation)

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-26	**Enter live**	Press Go live in Automation.	"Automation live mode" with LIVE badge; field animation shows Coggy, wires and sensor labels.			
TC-27	**Real time vs custom time**	Switch to Custom time, drag the slider to 6:05 AM, press play at 1 min/s.	Clock moves; sky and sun or moon change with time.			
TC-28	**Pump on by rule**	At 6:05 AM with dry soil.	Pump ON on the real pump, sprinklers animate, log entry with "Rule 1".			
TC-29	**Automation in the rain**	Keep 6:05 AM, splash the rain sensor (no rain stop rule).	Pump stays ON. Log shows a red "Pumping in the rain wastes water" note.			
TC-30	**Lights**	Cover the LDR.	Lights ON (real LED on GPIO 23), lamp glow in the field.			
TC-31	**Tested status**	Stay live for 15 s with readings.	"Tested" bar fills; toast "Automation is tested!".			
TC-32	**Exit live**	Press Exit live mode.	Pump and lights OFF; app moves to the AI tab.			

F. AI mode (5 days of training)

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-33	**Day 1 locked flow**	Try to open Day 2 before finishing Day 1.	Toast "Finish Day 1 first."			

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-34	**Save moments**	Set clock 6 AM, turn pump ON; set 9 AM, Save this moment; add evening moments.	Each moment appears in the diary with time, readings, pump and lights.			
TC-35	**Same time replaces**	Save twice at the same clock time.	Only one row for that time (latest wins).			
TC-36	**Day requirements**	Try Complete Day with fewer than 4 moments or one part of the day.	Button disabled; checklist shows what is missing.			
TC-37	**Rainy Day 3**	At watering time wet the rain sensor and keep pump OFF; save.	Row shows Raining and Pump OFF.			
TC-38	**Feed and train**	After Day 5, press Feed 5 days to Coggy.	Training animation, then "What Coggy learned" with rules, match % and a decision tree.			
TC-39	**Example diary**	On a fresh session press "Load an example diary", then Feed.	All 5 days fill; learned rules include "Pump OFF when … raining".			
TC-40	**Retrain**	Press Retrain from scratch and confirm.	Diary and AI results cleared; rules kept.			

G. Live mode (AI) and Compare

ID	Test	Steps	Expected result	Pass	Fail	Notes
TC-41	**AI in the rain**	Go live with AI. Custom time = watering time, soil dry, rain sensor wet.	AI turns pump OFF and explains why with % sure and closest memory.			
TC-42	**AI waters dry soil**	Dry the rain sensor, keep soil dry.	AI turns pump ON (if the diary showed watering dry soil).			
TC-43	**Compare locked**	Before both brains have 15 s of live testing, press Compare.	Toast says which brain still needs testing.			
TC-44	**Compare unlocked**	Test both, then open Compare.	Scores, "at a glance" table, challenge round, live tests table, bars and shared situations show.			
TC-45	**Copy results**	Press Copy results and paste in a document.	Text summary with scores and table rows.			
TC-46	**Refresh during live**	Refresh while live.	Live mode resumes after reconnecting Coggy; stats are kept.			